

Element C: Presentation and Justification of Solution Design

Requirements

Our requirements were drafted using both competition mandates and failure recovery strategies:

1. Fit through a 32" hoop
2. Maintain positive buoyancy at 3 feet and 6 feet depth (+0.25N net lift)
3. Use 6 thrusters (T0-T5) for full 3D maneuverability
4. Pull PVC cap within 10 seconds
5. Hold and release floatation devices with minimal drop error ($<10\%$)
6. Deliver rings and floating balls to home zone in <2 minutes

High-priority: buoyancy stability, thruster control reliability, and control accuracy.

These were validated through:

- Testing logs of thruster PWM consistency
- Depth trial comparisons (buoyancy drop vs. float correction)
- Direct feedback from Judge Ned Vorhees confirming polystyrene superiority
- Use of structured wiring and wire color tracking to reduce field maintenance errors

(Recreated digital sketch)

